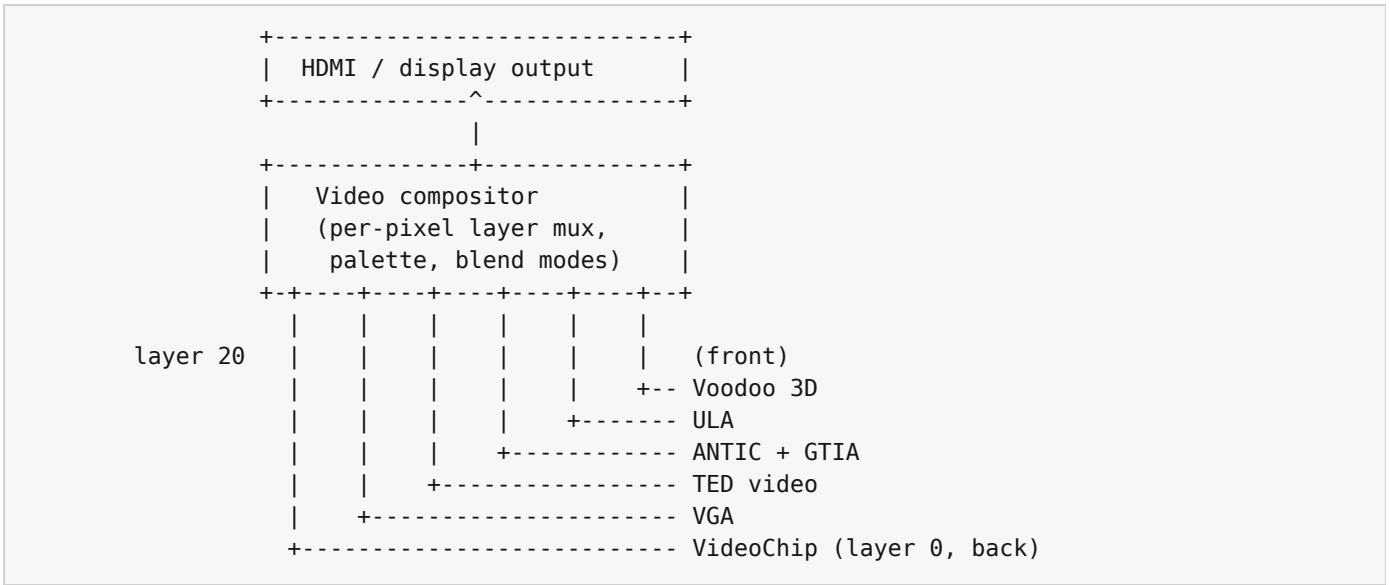


# Appendix K - Block Diagrams

Schematic-level layout of the three internal buses that connect the chips inside the Intuition Engine: the video compositor, the audio mixer, and the main system bus. Each diagram is drawn at the granularity a programmer needs to reason about cross-chip interactions, not at the granularity an engineer would use to fabricate the silicon.

## K.1 The video compositor

The compositor has six layers. Each layer comes from one of the six video engines. Layers are listed in back-to-front order; a higher layer covers the lower ones where it draws non-transparent pixels.



Each engine writes into its own framebuffer (in main VRAM or in its private aperture); the compositor reads all six on every output line and resolves them into a single stream of RGBA pixels.

All six sources are collected on the same 60 Hz frame cadence. A chip that has scanline work still completes that work in its own frame before the compositor resolves the stack.

## K.2 The audio mixer

The audio mixer has separate engine inputs, plus the SFX channels, and one stereo output stream.

```

SoundChip --+
PSG / AY --+
SN76489 ---+
SID -----+
SID2 -----+
SID3 -----+
POKEY -----+---> Mixer ---> Overdrive ---> Filter ---> Reverb ---> Output
TED audio -+      (sum,      (global)      (global)      (global)
AHX -----+      per-chip      voice)
MIDI/MUS --+      gain,
Live MIDI -+      per-chip
MOD -----+      mute)
WAV -----+
Paula DMA -+
          |
SFX ch 0-31 +

```

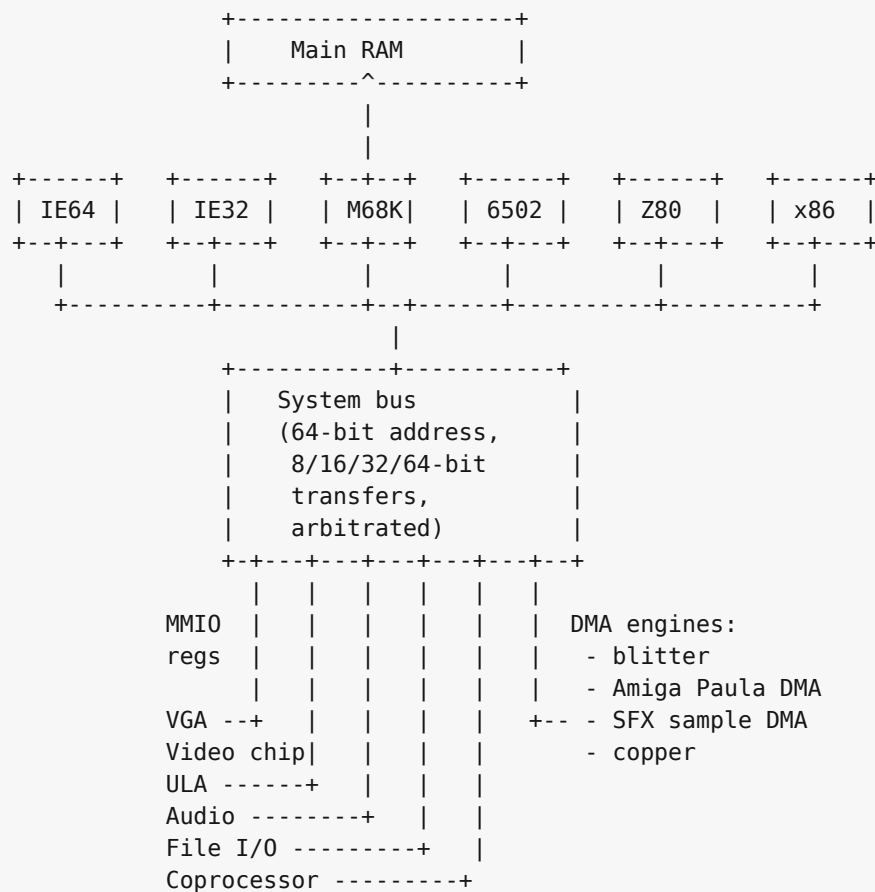
The SoundChip's own filter, the SID family's resonant filter, and the engine-internal effects all feed the mix before the global overdrive / filter / reverb stage; the global effects apply once per output sample to the summed signal.

MIDI/MUS and Live MIDI share the RawlandMini synth and voice pool. The diagram shows both control surfaces because one starts stored song data and the other streams immediate MIDI bytes.

## K.3 The system bus

---

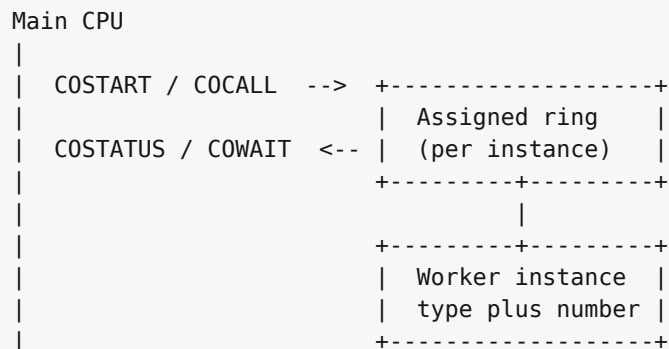
Every CPU and every device hangs off one shared 64-bit physical bus. The bus carries 64-bit physical addresses and supports 8, 16, 32, and 64-bit transfers through CPU and device adapters. It arbitrates simultaneous accesses on a round-robin basis; there is no priority encoding above the CPU / DMA distinction.



IE32, M68K, and x86 are 32-bit bus clients. The 8-bit CPUs (6502, Z80) reach the bus through an address translator that turns their 16-bit address space into selected low-window bus addresses, with the bank registers described in Chapters 27 and 28 selecting which translation applies.

## K.4 Coprocessor channels

The coprocessor block (Chapter 32) is a many-to-many channel between the main CPU and a pool of worker instances. Each running instance listens on its own ring buffer; the main CPU posts work items into that ring and reads the result back.



Six worker types share the protocol: an IE64 worker, an IE32 worker, a 6502 worker, a Z80 worker, an M68K worker, and an x86 worker. The main CPU is whichever one boots the machine; all the others are available as workers when not booted. M68K, x86, and IE64 provide instances 0 and 1; IE32, 6502, and Z80 provide instance 0 only.

The mailbox reserves twelve ring slots from \$790000 to \$792FFF. Each slot has a uniform \$400 stride, and its index is  $\text{cpuTypeIndex} * 2 + \text{instance}$ . The unused second slots for IE32, 6502, and Z80 remain reserved so every supported worker follows the same address rule.

## K.5 Bus translation for the small CPUs

The 6502 and Z80 see the same overall bus through a 16-bit-to- 64-bit translator. In practice their documented apertures target the low memory and MMIO window. The translator has two independent functions:

```
6502 / Z80 address (16-bit)
  |
  v
+---+-----+
| Decode page |
| - bank registers $F7xx? --> intercepted, never reach bus
| - MMIO mirror $F000-$FFF9? -> +$F0000 -> bus
| - $E000-$EFFF banked window-> + bank * 4 KB -> bus
| - everything else ----- > main RAM page
+---+-----+
  |
  v
bus address (64-bit, low-window value here)
```

The bank registers at \$F700-\$F705 (low/hi pairs for apertures 1, 2, 3) and \$F7F0 (aperture 0) are captured before the translator runs. A write to \$F700 does not reach \$F0700; it loads the low byte of bank register 1.